EXPERIMENT: - 3

AIM: Write 8085 Assembly language program to multiply two 8-bit numbers stored in memory location and store the 16-bit results into the memory.

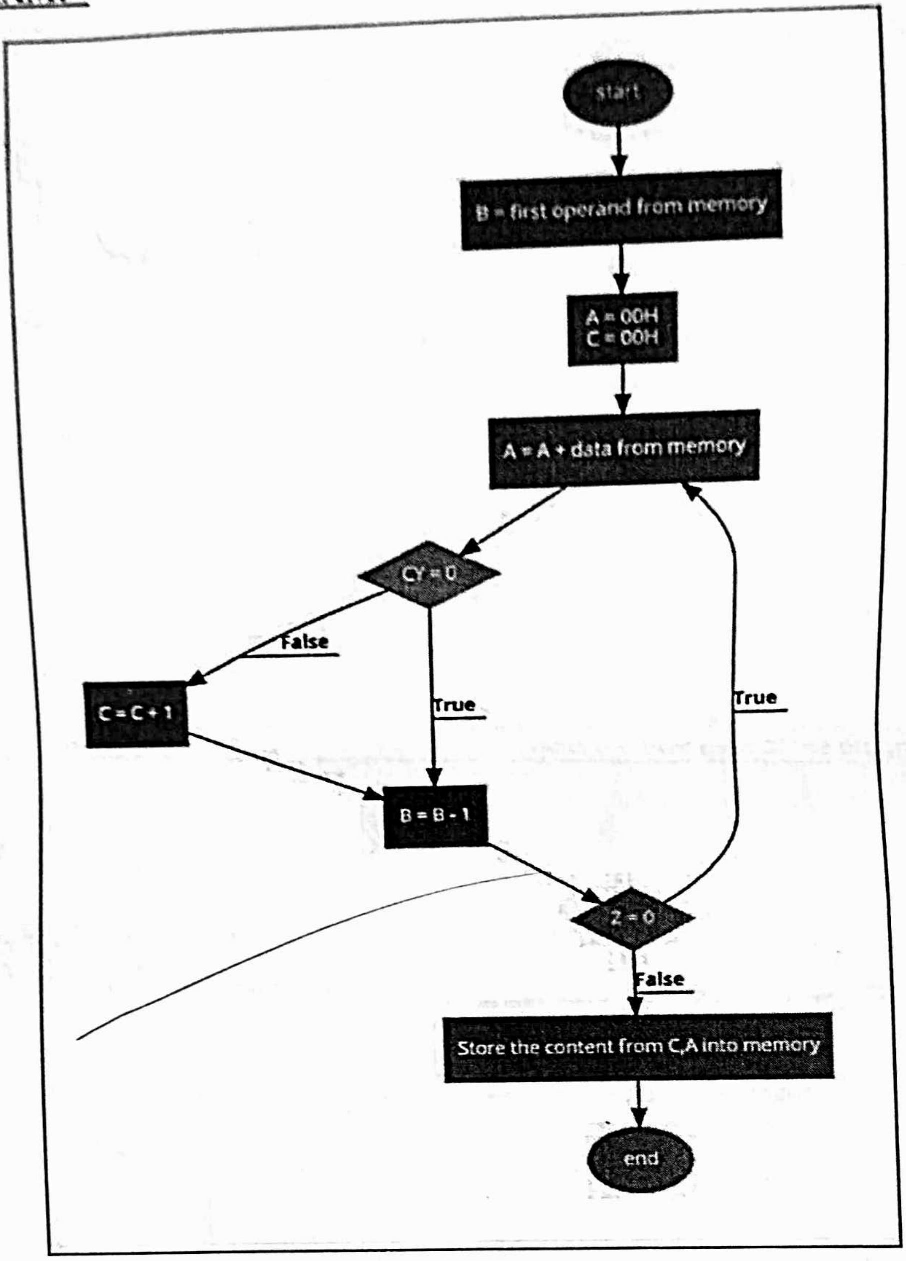
COMPONENTS REQUIRED: IC 7400, IC 7408, IC 7486, IC 7483, IC 7432, Patch Cords & IC Trainer Kit.

THEORY: The 8085 has no multiplication operation. To get the result of multiplication, we should use the repetitive addition method. After multiplying two 8-bit numbers it may generate 1-byte or 2-byte numbers, so we are using two registers to hold the result. We are saving the data at location 8000H and 8001H. The result is storing at location 8010H.

INPUT: -

ADDRESS	DATA
-	-
-	-
-	-
-	-
-	-
-	-
8000H	5
8001H	3
-	-
-	-
-	-
-	-
-	-
-	-
-	-

FLOW DIAGRAM: -



PROGRAM: -

ADDRESS	LABELS	MEMONICS	COMMENTS
		MVI C, 00	Moves immediate value 00 into CARRY C.
		MVI A, 00	Moves immediate value 00 into accumulator A.
8000		LXI H, 8000H	Loads immediate 16-bit value 8000H into register pair H-L, initializing a memory address.
		MOV B, M	Moves the value pointed to by the HL register pair into register B.
		INX H	Increments the HL register pair, pointing to the next memory location.
		MOV D, M	Moves the value pointed to by the HL register pair into register D.
	LOOP	ADD B	Adds the value in register B to accumulator A.
		JNC DOWN	Jumps to the label DOWN if there's no carry flag set after the addition.
	DOWN	DCR D	Decrements the value in register D.
		JNZ LOOP	Jumps back to the LOOP label if the value in register D is not zero.
8010		STA 8010H	Stores the value in accumulator A at memory address 8010H.
		HLT	Halts the execution of the program.

OUTPUT: -

ADDRESS	DATA
- -	- -
8010H	15
- -	- -

EXPERIMENT NO: - 10

AIM: To add and subtract two 8-bit numbers stored in memory and set various flags corresponding to the result.

APPARATUS: Battery, 8085 kit, power supply 220V.

INSTRUCTIONS USED FOR ADDITION:

Instruction MVI A, 8-bit data transfers the 8-bit number to the accumulator.

Instruction MVI A, 8-bit data transfers the 8-bit number to the accumulator.

The instruction ADD B adds the number in B to number in A.

This instruction i.e.: - STA 8500 H stores the result in the memory location.

HLT instruction ends the program

MEMORY ADDRESS	MNEMONICS	MACHINE CODES	COMMENTS
8000	MVI A, 20H	3E 20	LOADS THE NUMBER IN ACCUMULATOR.
8002	MVI B, 30 H	06 30	LOADS THE NUMBER IN REGISTER B.
8004	ADD B	80	ADDS THE CONTENTS STORED IN B TO THE ACC
8005	STA 8503 H	32 03 85	STORES THE RESULT AT 8503 H.
8008	HLT	76	STOP.

RESULTS:

8001-30H

8003-20H

8503: O/P=50H

INSTRUCTIONS USED FOR SUBTRACTION:

Instruction MVI A, 8-bit data transfers the 8-bit number to the accumulator.

Instruction MVI A, 8-bit data transfers the 8-bit number to the accumulator.

The instruction SUB B subtracts the number in B from number In A.

This instruction i.e.: - STA 8500 H stores the result in the memory location.

HLT instruction ends the program.

MEMORY ADDRESS	MNEMONICS	MACHINE CODES	COMMENTS
8000	MVI A, 30H	3E 20	LOADS THE NUMBER IN ACCUMULATOR.
8002	MVI B, 20 H	06 30	LOADS THE NUMBER IN REGISTER B.
8004	SUB B	90	ADDS THE CONTENTS STORED IN B TO THE ACC
8005	STA8503 H	32 03 85	STORES THE RESULT AT 8503 H.
8008	HLT	76	STOP

RESULTS:

8001-20H
8003-30H

8503: O/P=10H

EXPERIMENT NO: - 11

AIM: To implement various logical instructions involving comparing memory and register contents, logical operations - AND, OR, XOR and rotate operations.

THEORY:

Intel 8085 Instructions: - An instruction of a computer is a command given to the computer to perform a specified operation on given data. In microprocessor, the instruction set is the collection of the instructions that the microprocessor is designed to execute.

Logical Group: - The instructions in this group perform logical operations such as AND, OR compare, rotate, etc.

Instruction set	Explanation	States	Flags	Addressing	Machine cycles
ANA r [A] ←[A]∧[r]	AND register with accumulator	4	All	Register	1
ANA M [A] ←[A]∧[[H-]]	AND memory with accumulator	4	All	Register indirect	2
ANI data [A] ← [A] ∧ [data]	AND immediate data with accumulator	7	All	Immediate	2
ORA r [A] ←[A]∨[r]	OR-register with accumulator	4	All	Register	1
ORA M [A] ←[A]∨[[H-L]]	OR-memory with accumulator	7	All	Register indirect	2
ORI data [A] ← [A] ∨ [data]	OR immediate data with accumulator	7	All	Immediate	2
XRA r [A] ← [A]∀[r]	XOR register with accumulator	4	All	Register	1
XRA M [A] ← [A] ∀ [[H-L]]	XOR memory with accumulator	7	All	Register indirect	2
XRI data [A] ←[A] ∀ [data]	XOR immediate data with accumulator	7	All	Immediate	2
CMA [A] ←[A]	Complement the accumulator	4	None	Implicit	1
CMC [CS] ←[CS]	Complement the carry status	4	CS		1

STC [CS]← 1	Set carry status	4	CS		1
CMP r [A]-[r]	Compare register with accumulator	4	All	Register	1
CMP M [A] - [[H-L]]	Compare memory with accumulator	7	All	Register indirect	2
CPI data [A] - data	Compare immediate data with accumulator	7	All	Immediate	2
RLC [An+1] ←[An], [A0] ←[A7], [CS] ←[A7]	Rotate accumulator left	4	CS	Implicit	1
RRC [A7] ←[A0], [CS]←[A0], [An]←[An+1]	Rotate accumulator right		CS	Implicit	1
RAL [An+1] ←[An], [CS]←[A7], [A0] ←[CS]	Rotate accumulator left through carry		CS	Implicit	1
RAR [An] ←[An+1], [CS]←[A0], [A7] ←[CS]	Rotate accumulator right through carry		CS	Implicit	1